

AI-POWERED VISION-LANGUAGE IMAGE CAPTIONING SYSTEM USING BLIP

Submitted By: Mehakdeep Singh

Roll Number: 1024030827

1. Abstract

Image captioning is a computer vision task that combines image understanding and natural language generation to automatically describe the contents of an image. This project implements an AI-powered Vision-Language Image Captioning System using the BLIP (Bootstrapping Language-Image Pre-training) model. The system accepts an image as input and generates a meaningful textual description. The project demonstrates the practical application of Computer Vision and Natural Language Processing in real-world image understanding tasks.

2. Problem Statement

Traditional image processing systems can identify objects but often fail to provide meaningful descriptions of an entire scene. The objective of this project is to develop a Vision-Language system capable of understanding image content and generating human-readable captions automatically.

3. Objectives

- Develop a Vision-Language image captioning application.
- Generate automatic textual descriptions from images.
- Utilize a pre-trained BLIP model for image understanding.
- Provide a user-friendly interface for image upload and caption generation.
- Evaluate the effectiveness of generated captions on various test images.

4. Technologies Used

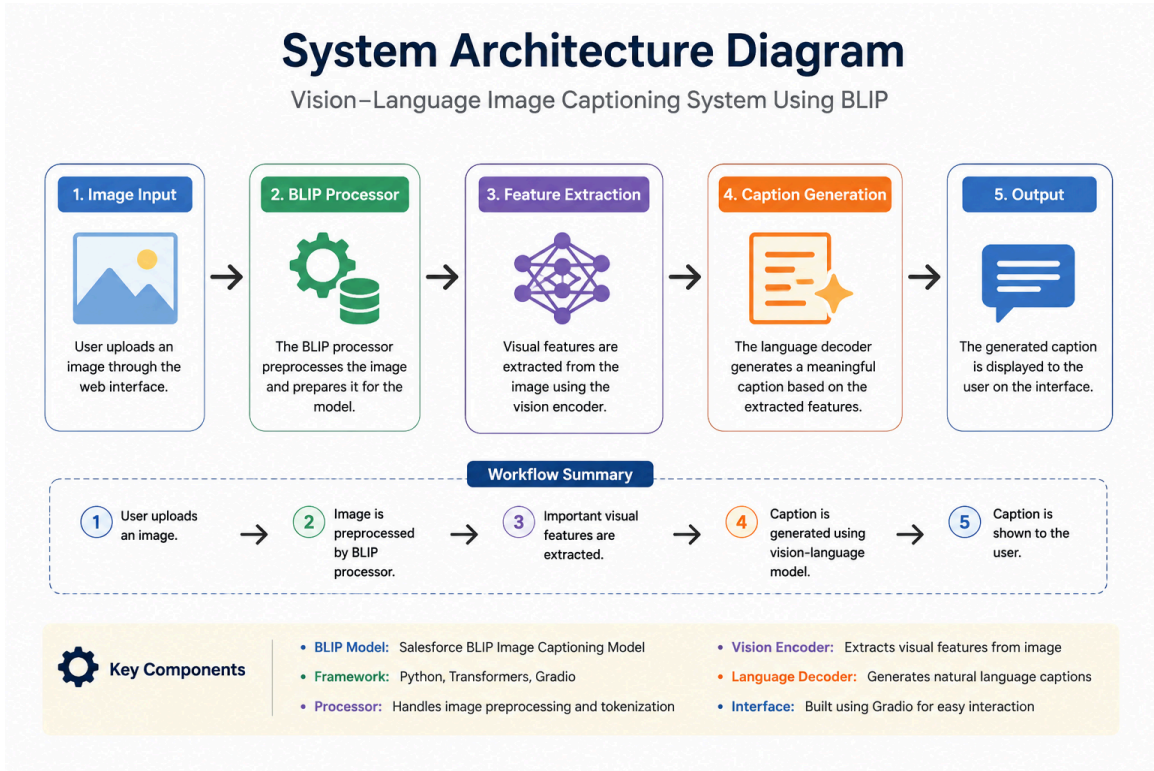
Python

Transformers

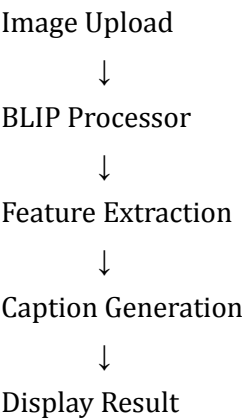
BLIP

Gradio
Pillow

5. System Architecture



6. Methodology



7. Dataset Information

The project uses publicly available images for testing and evaluation.

Reference Dataset:

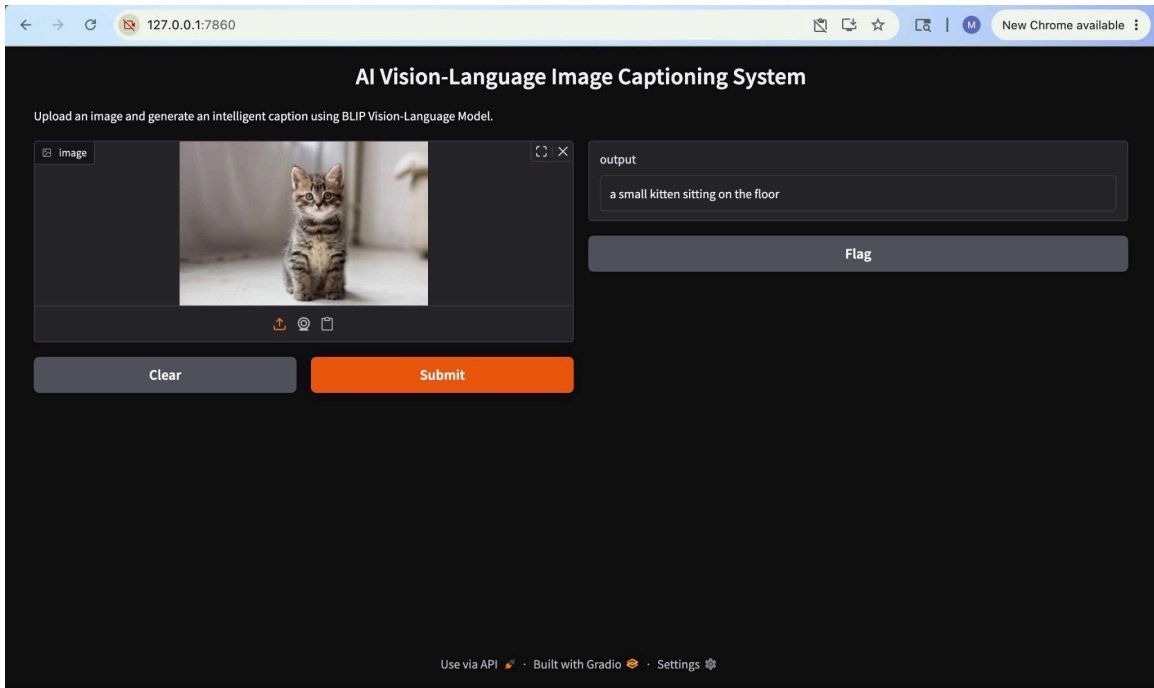
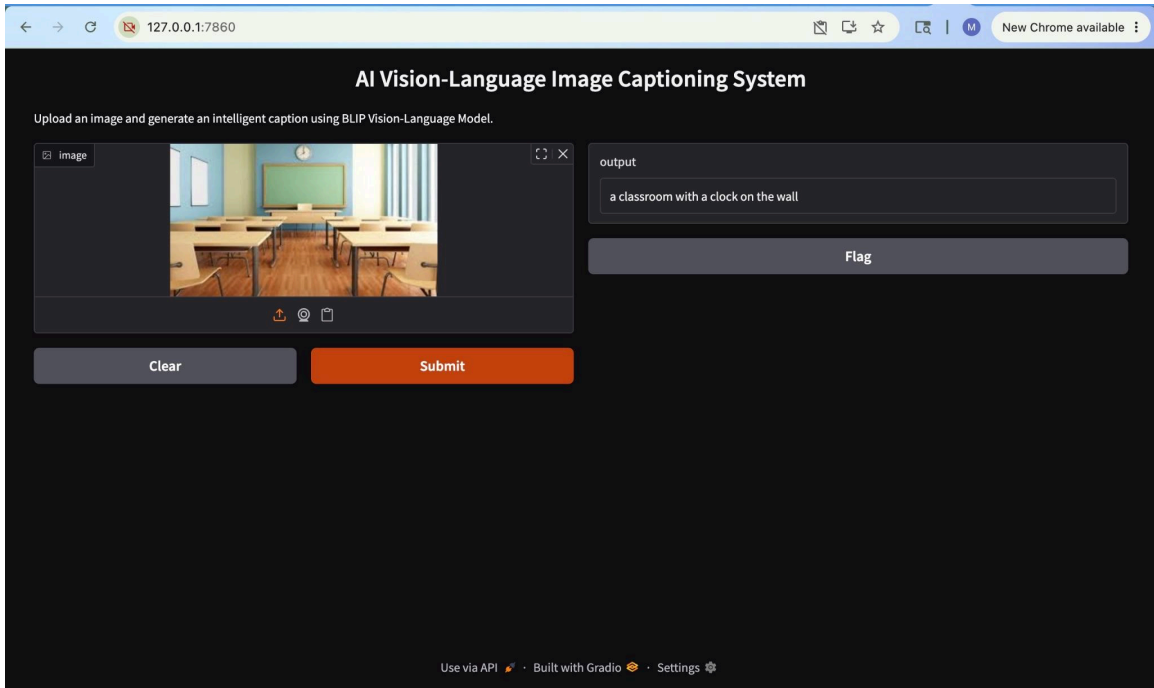
Flickr8k Dataset

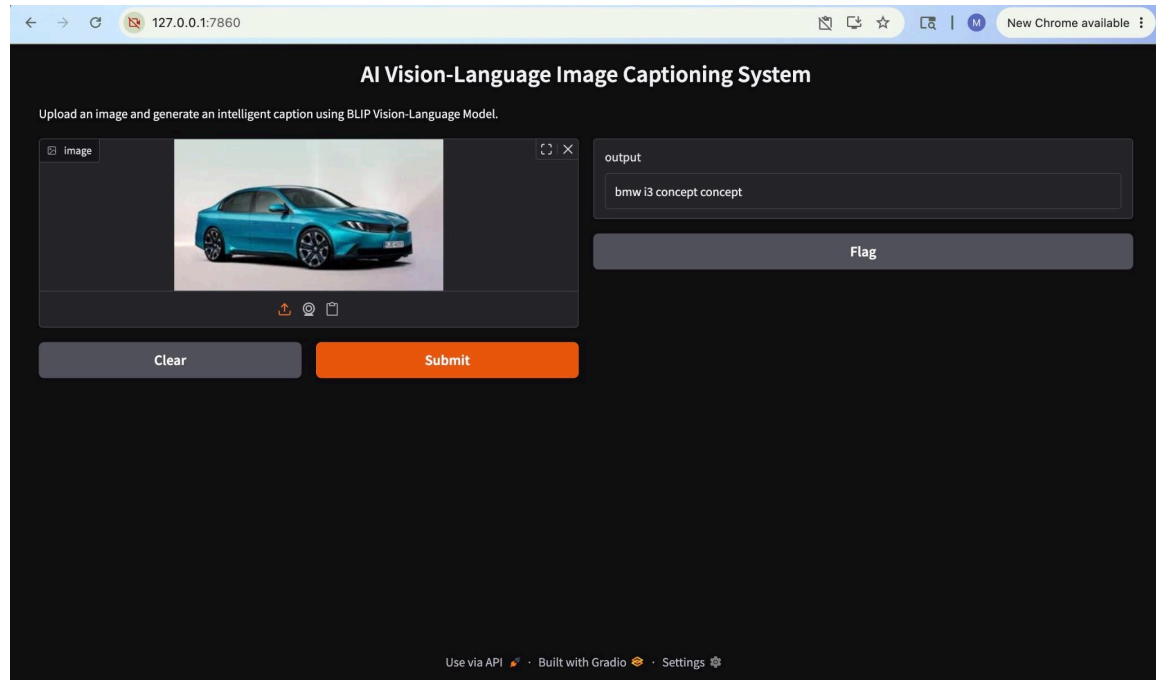
Dataset Link:

<https://www.kaggle.com/datasets/adityajn105/flickr8k>

The dataset contains thousands of real-world images paired with human-generated captions, making it suitable for evaluating image captioning systems.

8. Results and Performance Evaluation





9. Applications

The proposed system can be applied in:

- Assistive technologies for visually impaired individuals.
- Automated image indexing and retrieval systems.
- Social media content generation.
- Smart surveillance and scene understanding.
- Digital asset management systems.

10. Limitations

- Performance depends on image quality.
- Complex scenes may generate incomplete captions.
- Rare objects may not always be identified correctly.
- The system relies on a pre-trained model and is not fine-tuned for domain-specific applications.

11. Conclusion

The developed Vision-Language Image Captioning System successfully generates textual descriptions for uploaded images using the BLIP model. The project demonstrates how modern Vision-Language Models can combine computer vision and natural language processing to understand image content and generate meaningful captions. The system is efficient, user-friendly, and suitable for real-world image understanding applications.

12. Future Scope

Future enhancements may include:

- Visual Question Answering (VQA).
- Multi-language caption generation.
- Real-time video captioning.
- Fine-tuning on custom datasets.
- Integration with mobile and web platforms.

13. References

1. Salesforce BLIP Model Documentation.
2. Hugging Face Transformers Documentation.
3. Flickr8k Dataset.
4. Gradio Documentation.
5. Python Official Documentation.